

```

    },
    "ContactDetails": [
        {
            "ContactType1": "X",
            "Number": "X"
        },
        {
            "ContactType2": "X",
            "Number": "X"
        }
    ],
    "JobStatus": [X],
    "PAN": X
}

```

## Installing CouchDB

On Ubuntu and Debian Linux systems, use the following command:

```
$ sudo aptitude install couchdb
```

For Gentoo Linux, use the command below:

```
$ sudo emerge couchdb
```

The services can be started or stopped using the *init* scripts in all the distributions as follows:

```
$ /etc/init.d/couchdb start
```

The CouchDB installer is available for Windows from <http://couchdb.apache.org>. The CouchDB installed on the system can be executed in standalone mode as well as in service mode.

## Futon: A GUI administrator panel for CouchDB

Futon is the Web based GUI panel that is built for CouchDB. It provides the basic interface for a majority of the functions including creating, deleting, updating and viewing documents. It provides access to the configuration parameters and an interface for initiating replication (see Figures 3 and 4).

## CouchDB's interaction with Python

To interface Python with CouchDB, a specialised package called *couchdb* is used with the following main modules:

- *couchdb.client*: This is a client library for interfacing with CouchDB
- *couchdb.mapping*: This module provides the advanced mapping between JSON documents of CouchDB and Python objects
- *couchdb.view*: This module implements the view based server for the views written in Python

```
>>> import couchdb
```



Figure 3: Futon: The administration panel for CouchDB

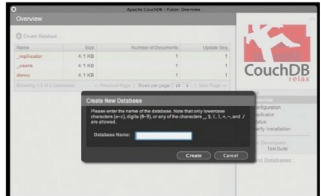


Figure 4: Database creation in Futon

```
>>> couch = couchdb.Server()
```

The above code creates the server object. The default URL is *localhost:5984*.

On live servers, the code will be as follows:

```
>>> couch = couchdb.Server('http://www.mybigcloudportal.com:5984/')
>>> mydb = couch.create('testDB')
```

The above code creates a new database.

```
>>> mydb = couch['mynosqlldb']
```

The above code is to use the existing database.

```
>>> mydoc = {'Country': 'India'}
```

After selecting a database, the above code is used to create a document and insert it into the database.

```
>>> mydb.save(mydoc)
```

The *save()* method shown above returns ID and the 'rev' for the currently created document.

```
>>> mydb.delete(doc)
```

```
>>> couch.delete('testDB')
```